Homework 2
Out: February 18th, 2026, Wednesday -- Due: March 16th, 2026, Monday, 11:59pm
EC327 Introduction to Software Engineering – Spring 2026

Total: 160 points

Submission

- Write your answers to each question area (e.g. Q1, Q2, Q3) clearly in your favorite text editor or word processor and submit a **pdf** file for each question **AREA** to Gradescope. In this case that means you will have 3 pdfs:
    - Username_Q1.pdf
    - Username_Q2.pdf
    - Username_Q3.pdf
- Within each PDF, each sub-question should be on its own page. This will make grading with Gradescope much easier for us. For example:
    - Username_Q1.pdf should have 15 pages for the 15 questions in its section.
    - If you have questions, please ask via Piazza.
- Submit your assignment to Gradescope
    - There will be three different submissions: One for Q1, one for Q2, and one for Q3

Failure to follow submission process will cost you points.

You can use the internet, books, etc. to learn more about how to answer these questions. You can also re-type the code in your coding environment and actually run it. **What you CAN'T DO is just put the question into AI and ask it for the answer.** Again, ask if you have questions. The point of this type of homework is to emphasize the academic part of the class. These are the types of questions you could get at a coding interview!

Q1. "Book Problems" [75 points]
5pts each; 15 total problems

A.

List all keywords in the following program.

```
#include <iostream>
using namespace std;

int main ()
{
        int x = 0;
        int y = 1;
        cout << x << y;
        return 0;
}
```

B.

Find all non-keyword identifiers in the following program.

```
#include <iostream>
using namespace std;

int main ()
{
    int x = 4;
    int y = 22;
    cout << x << y;
    return 0;
}
```

C.

Which of the following variables are correctly declared?

a. `char letter = 'A';`
b. `int first, second;`
c. `double average, tax = 8.5;`
d. `long double 3x;`

D.

What will be output from the following program segment?

```
double x = -909.245;
cout << fixed << setprecision (2) << x << endl;
cout << fixed << setw (15) << setprecision (3) << setfill ('*') << x;
```

E.

If originally x = 4, y = 0, and z = 2, what are the values of x, y, and z after executing the following code?

```
if ( z == 0 || y != 0 )
{
    if ( z <= 2 )
    {
        z = 4;
    }
}
else
{
    y = 5;
    z = y + x;
}
```

F.

Rewrite the following code fragment using one *switch* statement.

```
if (ch == 'A' || ch == 'a')
    countA++;
else if (ch == 'E' || ch == 'e')
    countE++;
else if (ch == 'I' || ch == 'i')
    countI++;
else
    cout << "Error--Not A, E, or I " << endl;
```

G.

How many times is the body of the following loop executed? What is the value of x after the loop termination?

```
int x = 15;
do
{
    x -= 2;
    if (x < 9)
    break;
} while (true);
```

H.

Change the following *do-while* loop to a *while* loop.

```
int x = 10;
do
{
    cout << x << end;
    x++;
} while (true);
```

I.

Given the declaration of the following two functions, determine if they are overloaded versions of each other.

```
int fun (int x, int y, int z);
float fun (int a, int b);
```

J.

Find any errors in the following function definition:

```
void one ()
{
        cout << "In One" << endl;
        void two ()
        {
                cout << "In Two" << endl;
                return;
        }
        return;
}
```

K.

What is the output from the following code segment?

```
int arr [10] = {0, 1, 2, 3, 4, 5, 6, 7, 8, 9};
for (int i = 0; i < 5; i++)
{
        arr [i ] = arr [i * 2 + 1];
        cout << arr [i] << endl;
}
```

L.

What is wrong with the following lines of code?

```
int x = 1000;
int& y = 2000;
```

M.

What is wrong with the following lines of code?

```
const int x = 100;
double& y = x;
```

N.

Given the following lines of code, show what would be printed.

```
int x = 100;
int* y = &x;
int* z = &x;
cout << x << " " << *y << " " << *z;
```

O.

What is created in the heap from the following statements? Draw a picture to show the object in the heap.

```
int** arr = new int* [3];
for (int i = 0; i < 3; i++)
{
    arr[i] = new int [5];
}
```

Q2. Pointer Practice [50 points]

Mark the following as true or false. If false provide a 1-2 sentence explanation.

PUT ALL OF THESE ON ONE PAGE! You do NOT need a separate page for parts a-h

a. In C++, pointer is a reserved word.

b. In C++, pointer variables are declared using the word “pointer”.

c. The statement **delete p**; deallocates the pointer p.

d. The statement **delete p**; deallocates the dynamic memory that is pointed to by p.

e. Given the declaration:

```
int list[10];
int *p;
```

The statement:

```
p = list;
```

is valid in C++

f. Given the declaration:

```
int *p;
```

The statement:

```
p = new int[50];
```

dynamically allocates an array of 50 components of type int and p contains the base address of the array.

g. The address of operator (&) returns the address and the value of its operand.

h. If p is a pointer variable, then the statement p = p * 2; is valid in C++;

Answer the following five questions related to code output (Q2 i through m). **Each of these answers SHOULD be on a separate page.**

i. What is the output of the following codes snippet:

```
int num1;
int num2;
int *p = &num1;
p = &num2;
*p = 25;
num1 = num2 + 6;
p = &num1;
num2 = 73;
*p = 47;
cout << *p << " " << num1 << " " << num2 << endl;
```

j. What is the output of the following code snippet:

```
int *length;
int *width;
length = new int;
*length = 5;
width = length;
length = new int;
*length = 2 * (*width);
cout << *length << " " << *width << " " << (*length) * (*width)
     << endl;
```

k. What is the output of the following code snippet:

```
int num;
int *listPtr;
int *temp;
listPtr = new int[5];
num = 8;
temp = listPtr;

for (int j = 0; j < 5; j++)
{
    *listPtr = num;
    num = num + 2;
    listPtr++;
}

listPtr = temp;

for (int k = 0; k < 5; k++)
{
    *temp = *temp + 3;
    temp++;
}

for (int k = 0; k < 5; k++)
{
    cout << *listPtr << " ";
    listPtr ++;
}
cout << endl;
```

l) What is the output of the following code snippet:

```
int *myList = new int[5];
int *yourList = new int[10];

myList[0] = 3;

for (int i = 1; i < 5; i++)
    myList[i] = myList[i - 1] + i;

for (int i = 0; i < 5; i++)
{
    yourList[i] = myList[i] + 4;
    yourList[i + 5] = myList[4 - i] - 3;
}

cout << "myList: ";
for (int i = 0; i < 5; i++)
    cout << myList[i] << " ";
cout << endl;

cout << "yourList: ";
for (int i = 0; i < 10; i++)
    cout << yourList[i] << " ";

cout << endl;
```

m) Consider this code snippet:

```
int count = 7;
int *pcount = &count;
cout << pcount << endl; //prints the memory address 0x7fff560b
```

For each statement in the table, write the value of count, *pcount, and pcount. Assume that after each statement, all values are reinitialized.

You can have unknown as your answer

	count	*pcount	pcount
count += 2;			
*pcount++;			
*(++pcount);			
--*pcount+*pcount;			

Q3. General Questions [35 points]

a. List 6 high level programming languages, roughly the year they were created, and an example of what they are good for.

b. How did compilers come about if you need a compiler to compile a program in the first place?

c. Who are: (2 sentences each)

   a. Dennis Richie
   b. Alan Turing
   c. James Gosling
   d. John Backus
   e. Steve Wozniak
   f. Pick a computer scientist/engineer of your choice

d. What is a “goto statement” in a programming language and why are they discouraged?

e. Produce the Assembly code for one of your questions from PA1 using:

g++ -S <filename.cpp>

Turn in some of the assembly code. What do you find most interesting about this?

f. Produce the C++ code after the preprocessor runs for one of your questions from PA1 using:

g++ - E <filename.cpp> > newOutput.cpp

Turn in some of the code. What do you find most interesting about this?

g. What is one thing you want to know more about regarding computing (most interesting will be shared with in class; maybe even answered).